

Jharkhand University of Technology, Ranchi**B.Tech. 1st Semester Examination, 2023 (NEP)****Subject : Engineering Chemistry****Subject Code : BSC02****Time Allowed : 3 Hours****Full Marks : 70**

Candidates are requested to give the answers in their own words as far as practicable.

Answer any five questions in which Question No. 1 is compulsory.

All questions are of equal value.

1. Answer the following questions :

2×7=14

(i) Which of the following wavelength fall in IR spectrum?

(a) 2000 nm

☒ (b) 110 nm

(c) 800 nm

(d) 10 cm

(ii) Identification of an organic compound that can established by using

☒ (a) IR spectroscopy

(b) UV spectroscopy

(c) Any of these

(d) None of these

(iii) The metal at the top of electrochemical series is

☒ (a) More active

(b) Less active

(c) Most stable

(d) More noble

(iv) Difficult to monitor and very dangerous form of corrosion—

☒ (a) Pitting

☒ (b) Galvanic

(c) Crevice

☒ (d) Stress

(v) The nanomaterials have the following characteristics—

Which do not exist in the corresponding bulk materials?

(a) High surface energy

☒ (b) Greater imperfection

☒ (c) Smaller number of surface atom

(d) None of these

(vi) Which of the following nanomaterials is utilized as a solid-fuel in rocket propulsion?

☒ (a) Nano aluminium

(b) Nano zirconium

(c) Nano silicate

(d) All of these

(vii) The capacity of a cell is measured in

☒ (a) Ampere - hour

(b) Amperes

(c) Watts

(d) Watt - hours

X 2. (a) Discuss about following:

• (i) Quality control in engineering context

• (ii) Role of refractories in engineering

• (b) Write the principle of UV spectroscopy. Discuss various electronic transitions. (3.5×2)+7

3. (a) State and explain electrochemical theory of corrosion.

• (b) Discuss about various corrosion prevention techniques. 7+7

4. (a) Define battery. Differentiate between electrolytic cell and electrochemical cell.

• (b) Explain principle, working and application of fuel cells. 7+7

5. (a) Define nanomaterials. Explain the mechanical and magnetic properties of nanomaterials.

• (b) Explain various importance of nanomaterials. 7+7

6. (a) Explain top-down and bottom-up approach of nanomaterials.

(b) Discuss about green hydrogen technology.

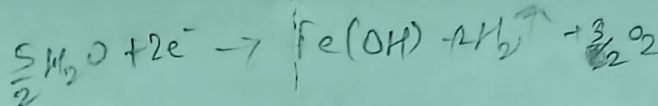
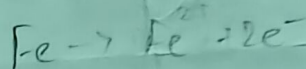
X 7. Write notes on following:

• (a) chromophores

• (b) auxochromes

• (c) HPLC

• (d) Galvanic corrosion



A.	C
color	Colorless
(0.61)	(0.61)

7+7

3.5×4